

NOBEOKA, Japan

WMO: 478220

Lat: 32.5819N

Lon: 131.6581E

Elev: 69

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.2	32.2	15.5	12.1	39.5	18.1	13.7	40.2	23.5	46.2	21.2	45.8	4.8	270	0.346

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	90.8	78.8	89.0	78.5	87.4	78.0	80.7	87.1	79.9	86.0	79.1	84.9	7.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	151.4	83.8	78.3	147.2	83.1	77.4	143.0	82.5	44.4	87.2	43.5	86.1	42.7	85.0	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.4	16.1	13.3	DB	26.1	95.0	1.7	1.6	24.9	96.1	23.9	97.0	22.9	97.9	21.6	99.1	
			WB	23.8	82.3	2.0	1.0	22.4	83.0	21.2	83.5	20.0	84.1	18.6	84.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	44.6	46.7	52.2	60.0	67.5	72.8	80.0	81.4	76.2	67.4	57.5	48.1
	DBStd	13.48	4.81	5.84	5.54	4.98	3.63	3.95	3.07	2.41	4.00	4.91	5.89	5.46
	HDD50	457	181	125	39	2	0	0	0	0	0	0	7	103
	HDD65	2506	634	512	397	159	17	0	0	0	0	30	233	524
	CDD50	5187	12	33	108	302	541	685	930	974	786	540	231	45
	CDD65	1760	0	0	0	10	93	235	465	509	336	105	7	0
	CDH74	13821	0	0	3	29	341	1164	4286	5133	2412	441	12	0
	CDH80	4140	0	0	0	1	23	211	1405	1855	605	40	0	0
Wind	WSAvg	5.2	6.1	6.1	6.3	5.4	4.7	3.8	4.2	4.7	4.9	5.2	5.2	5.9
Precipitation	PrecAvg	93.5	2.5	3.2	6.2	7.8	9.7	15.2	11.4	10.6	13.0	7.2	4.2	2.4
	PrecMax	133.2	6.2	6.8	15.1	14.1	28.3	27.3	33.9	25.4	24.4	22.8	12.5	6.2
	PrecMin	59.6	0.6	0.9	2.5	2.2	3.9	5.2	1.3	1.6	2.6	0.3	0.7	0.0
	PrecStd	18.4	1.6	1.9	3.0	3.3	5.2	5.5	8.2	5.9	6.3	6.7	2.9	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.7	68.1	72.4	77.4	82.7	89.9	93.1	93.5	89.8	83.7	75.5	66.7
		MCWB	55.6	59.8	60.3	66.0	67.9	77.7	78.4	79.0	77.6	73.9	65.8	57.2
	2%	DB	59.3	64.1	68.3	74.1	79.6	85.0	90.3	91.3	87.3	80.8	72.2	63.5
		MCWB	50.7	57.0	57.3	62.7	66.7	75.6	78.7	79.1	76.5	70.5	63.0	55.2
	5%	DB	56.6	60.3	65.4	71.8	77.4	81.9	88.6	89.6	85.4	78.5	70.0	60.8
		MCWB	48.6	52.6	56.1	61.3	66.0	73.2	78.5	78.9	75.6	68.7	61.9	52.7
	10%	DB	54.1	57.2	62.6	69.8	75.7	79.8	86.9	88.0	83.6	76.4	67.5	58.0
		MCWB	46.5	49.6	54.3	60.4	65.6	72.1	78.1	78.6	74.5	67.3	60.2	51.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.7	62.7	64.0	70.0	74.2	79.1	81.7	82.1	80.5	77.1	69.3	61.8
		MCDB	61.6	66.0	69.5	72.4	77.9	85.3	88.1	88.6	86.0	80.3	72.0	63.4
	2%	WB	53.7	58.7	61.0	67.2	72.0	77.5	80.5	81.0	79.0	74.5	66.5	57.6
		MCDB	56.6	62.7	64.7	70.4	75.0	82.7	86.8	87.4	84.0	78.1	69.3	60.9
	5%	WB	50.2	55.0	58.8	64.9	70.0	75.9	79.6	80.2	77.9	72.0	64.2	54.9
		MCDB	54.9	58.5	63.0	69.0	73.2	80.3	85.7	86.4	82.6	75.7	67.8	58.7
	10%	WB	47.4	51.2	56.2	62.9	68.3	74.4	78.8	79.5	76.8	69.6	61.9	52.1
		MCDB	52.9	55.8	61.1	67.5	72.4	78.0	84.7	85.5	81.2	73.8	65.8	56.8
Mean Daily Temperature Range	5% DB	MDBR	17.1	17.3	17.5	17.3	15.7	11.8	11.9	12.9	13.6	15.8	16.7	17.1
		MCDBR	20.0	19.7	20.1	19.5	18.6	15.3	15.3	15.5	15.3	17.4	17.6	19.2
		MCWBR	13.2	13.1	12.8	11.9	10.4	7.5	6.7	6.7	7.3	9.2	10.6	12.1
	5% WB	MCDBR	17.2	17.0	16.5	15.2	13.5	12.5	13.2	13.4	12.3	12.2	13.6	15.5
		MCWBR	13.5	13.5	12.8	11.1	9.6	6.9	6.5	6.4	6.9	7.9	9.8	12.0
Clear-Sky Solar Irradiance	taub	0.418	0.464	0.529	0.552	0.563	0.570	0.558	0.541	0.516	0.478	0.432	0.406	
	taud	2.113	2.020	1.864	1.861	1.901	1.963	2.034	2.074	2.104	2.123	2.168	2.167	
	Ebn at Noon	245	245	239	240	239	236	238	240	240	239	239	241	
	Edh at Noon	41	49	62	64	62	59	54	51	48	44	38	36	
All-Sky Solar Radiation	RadAvg	849	1023	1324	1524	1585	1273	1570	1584	1302	1102	870	770	
	RadStd	70	89	86	163	185	204	198	141	148	125	92	54	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page